ADVENTURES IN BIG MATH · 3RD GRADE MATH

MISSION 06 OF 08 · WORKSHEET PACK

Geometry Quest

Every graded set, every week — 5 weeks, 25 worksheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

WEEK 5

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

NAME __________

DATE __________

SCORE ____ / 11

What an Angle Is

Every answer is a number of degrees, or a count. Type digits only.

Warm-Up

6 ITEMS

Degrees in a right angle	Degrees in a straight line	Degrees all the way round
Right angles in a full turn	Degrees in half a right angle	Right angles in a straight line

Core

GRADED · SHOW YOUR WORKING

1. On a straight line: 130° and ?° (180 - 130)
2. Round a point: 90°, 120° and ?° (360 - 210)
3. Three angles of a triangle: 60°, 70° and ?° (They add to 180)
4. An angle of 200° — how far past a straight line? (200 - 180)
5. Two equal angles on a straight line — each is ?° (180 shared evenly)

Challenge

NOT GRADED

C1. Angles in a square, added up (Four right angles)

C2. Turn 90° four times — total degrees (All the way round)

3RD GRADE · MISSION 06 · WEEK 1 · SET 1.2

Perimeter & Area

Perimeter is the fence. Area is the grass. Read which one is being asked for.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

Perimeter of a 5 by 3 rectangle	Area of a 5 by 3 rectangle	Perimeter of a square with side 6
Area of a square with side 6	Perimeter of a 10 by 2 rectangle	Area of a 10 by 2 rectangle

Core GRADED · SHOW YOUR WORKING

1. Area 48, one side 6 — the other side (48 ÷ 6)
2. Perimeter 30, one side 9 — the other side (9 + 9 is 18, and 12 is left for two sides)
3. Area of a 12 by 7 rectangle (84 square units)
4. Perimeter of a 12 by 7 rectangle (12 + 12 + 7 + 7)
5. L-shape: a 6 by 4 block plus a 3 by 2 block — total area (24 + 6)

Challenge NOT GRADED

C1. Biggest area with perimeter 24 (whole sides) (The 6 by 6 square)

C2. Smallest area with perimeter 24 (whole sides) (1 by 11)

The Coordinate Plane

Across first, then up. (3, 5) is a different place from (5, 3).

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

In (3, 5), the across number	In (3, 5), the up number	In (7, 2), the across number
In (0, 4), the across number	In (6, 6), the up number	In (9, 1), the up number

Core GRADED · SHOW YOUR WORKING

1. Distance from (2, 3) to (7, 3) (Only the across number changed) ....
2. Distance from (4, 1) to (4, 8) (8 - 1) ....
3. Rectangle at (1,1), (6,1), (6,4), (1,4) — its perimeter (5 wide and 3 tall) ....
4. That same rectangle — its area (5 × 3) ....
5. Start at (2,2), go 4 across and 3 up — the up number now (2 + 3) ....

Challenge NOT GRADED

C1. Square with corners (0,0), (5,0), (5,5) — the fourth corner's up number (It sits at (0, 5))

....

C2. Distance from (1,2) to (1,9) plus (3,4) to (8,4) (7 + 5)

....

Sorting Shapes & Symmetry

Sort by what is true about a shape, not by what it looks like.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

Sides on a hexagon	Sides on a pentagon	Right angles in a rectangle
Equal sides on an equilateral triangle	Sides on an octagon	Pairs of parallel sides in a rectangle

Core GRADED · SHOW YOUR WORKING

1. Lines of symmetry in a square (Two through the middle, two through the corners)
2. Lines of symmetry in a rectangle (not a square) (The corner folds don't match)
3. Lines of symmetry in an equilateral triangle (One from each corner)
4. Lines of symmetry in a circle — type 0 if too many to count (Infinitely many, so type 0)
5. Degrees in a quarter turn (A right angle)

Challenge NOT GRADED

C1. Lines of symmetry in a regular hexagon (One per pair of opposite corners and sides)

C2. Sides of a shape whose angles add to 360 (Any quadrilateral)

Angle Guess & Map a Planet

Enrichment day. Estimate before you measure, every single time.

NAME __________

DATE __________

ENRICHMENT

★ Challenge

NOT GRADED

C1. Angle on a straight line next to 47° (180 - 47)

C2. Angles round a point: 100°, 130° and ?° (360 - 230)

C3. Rectangle perimeter 20 with the largest area — that area (The 5 by 5 square)

C4. Rectangle area 36 with the smallest perimeter — that perimeter (6 by 6)

C5. Distance from (3,7) to (11,7) (11 - 3)

C6. A shape with 4 lines of symmetry and 4 equal sides — its angle count (It's a square)

C7. Turn 45° eight times — total degrees (45 × 8)

C8. Interior angles of a triangle, added (Always)

Week 1 Answers

Mission 06 · all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

1.1 · What an Angle Is

OUT OF 11

Warm-Up · 90 | 180 | 360 | 4 | 45 | 2

Core · 50 | 150 | 50 | 20 | 90

Challenge · 360 | 360

1.2 · Perimeter & Area

OUT OF 11

Warm-Up · 16 | 15 | 24 | 36 | 24 | 20

Core · 8 | 6 | 84 | 38 | 30

Challenge · 36 | 11

1.3 · The Coordinate Plane

OUT OF 11

Warm-Up · 3 | 5 | 7 | 0 | 6 | 1

Core · 5 | 7 | 16 | 15 | 5

Challenge · 5 | 12

1.4 · Sorting Shapes & Symmetry

OUT OF 11

Warm-Up · 6 | 5 | 4 | 3 | 8 | 2

Core · 4 | 2 | 3 | 0 | 90

Challenge · 6 | 4

Fri · Angle Guess & Map a Planet

ENRICHMENT

Challenge · 133 | 130 | 25 | 24 | 8 | 4 | 360 | 180

Perimeter

The distance all the way round. Add every side, or be clever about it.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up
6 ITEMS

Perimeter of a 4 by 3 rectangle	Perimeter of a 5 by 5 square	Perimeter of a 10 by 2 rectangle
Perimeter of a 6 by 6 square	Perimeter of a 7 by 1 rectangle	Perimeter of a 3 by 3 square

Core
GRADED · SHOW YOUR WORKING

1. Perimeter of a 12 by 8 rectangle
2. A square of perimeter 36 — one side
3. A rectangle of perimeter 20, one side 6 — the other
4. Perimeter of a 15 by 9 rectangle
5. A square of perimeter 100 — one side

Challenge
NOT GRADED

C1. A rectangle of perimeter 30 with whole sides — how many different ones

C2. Perimeter of an L made from a 5 by 5 square with a 2 by 2 corner removed

Area

Length times width — and count the squares if you doubt it.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up
6 ITEMS

Area of a 4 by 3 rectangle	Area of a 5 by 5 square	Area of a 10 by 2 rectangle
Area of a 6 by 6 square	Area of a 7 by 1 rectangle	Area of a 8 by 3 rectangle

Core
GRADED · SHOW YOUR WORKING

1. Area of a 12 by 8 rectangle ....
2. Area 48, one side 6 — the other ....
3. Area of a 15 by 9 rectangle ....
4. A square of area 49 — one side ....
5. Area 144 with equal sides — one side ....

Challenge
NOT GRADED

C1. Area of an L: a 6 by 6 square minus a 2 by 3 corner

C2. Double both sides of a 4 by 5 rectangle — the new area

Same Fence, Different Field

Perimeter fixed, area wildly different. That is the Big Question.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

A 1 by 11 rectangle — perimeter	That rectangle — area	A 6 by 6 square — perimeter
That square — area	A 2 by 10 rectangle — area	A 4 by 8 rectangle — area

Core GRADED · SHOW YOUR WORKING

1. With 24 m of fence, the biggest whole-number area ....
2. With 20 m of fence, the biggest area ....
3. With 16 m of fence, the biggest area ....
4. With 24 m of fence, the smallest area using whole sides ....
5. With 30 m of fence, the biggest whole-number area ....

Challenge NOT GRADED

C1. The shape that always gives the biggest area for a fixed perimeter — type square or circle

....

C2. Among rectangles with perimeter 40, the biggest area

....

NAME __________
DATE __________
SCORE ____ / 11

Area of Compound Shapes

Split it into rectangles, do each, add them up.

Warm-Up

6 ITEMS

A 3 by 2 plus a 3 by 2 — total area	A 4 by 4 plus a 2 by 2	A 5 by 3 plus a 5 by 1
A 6 by 2 plus a 3 by 2	A 10 by 1 plus a 5 by 2	A 7 by 3 plus a 1 by 3

Core

GRADED · SHOW YOUR WORKING

1. An 8 by 6 with a 2 by 3 removed — area
2. A 10 by 10 with a 4 by 4 removed
3. An L of a 9 by 4 and a 5 by 3
4. A 12 by 5 with a 3 by 5 removed
5. A T of a 8 by 2 and a 2 by 6

Challenge

NOT GRADED

C1. A 20 by 15 room with a 4 by 5 alcove added — total area

C2. A 12 by 12 with a 12 by 3 strip removed

NAME __________

DATE __________

SCORE ____ / 4

The Biggest Pen

24 metres of fence, and a wall to build against. Now what?

Warm-Up

2 ITEMS

With 24 m and no wall, the biggest area

Sides you must fence with a wall behind you

Core

GRADED · SHOW YOUR WORKING

1. 24 m against a wall, 8 m deep each side — the width

2. That pen's area

Challenge

NOT GRADED

C1. 24 m against a wall, 6 m deep each side — the area

C2. The biggest area for 24 m against a wall

C3. How much bigger than the no-wall best

Week 2 Answers

Mission 06 · all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

2.1 · Perimeter

OUT OF 11

Warm-Up · 14 | 20 | 24 | 24 | 16 | 12

Core · 40 | 9 | 4 | 48 | 25

Challenge · 7 | 20

2.2 · Area

OUT OF 11

Warm-Up · 12 | 25 | 20 | 36 | 7 | 24

Core · 96 | 8 | 135 | 7 | 12

Challenge · 30 | 80

2.3 · Same Fence, Different Field

OUT OF 11

Warm-Up · 24 | 11 | 24 | 36 | 20 | 32

Core · 36 | 25 | 16 | 11 | 56

Challenge · circle | 100

2.4 · Area of Compound Shapes

OUT OF 11

Warm-Up · 12 | 20 | 20 | 18 | 20 | 24

Core · 42 | 84 | 51 | 45 | 28

Challenge · 320 | 108

Fri · The Biggest Pen

OUT OF 4

Warm-Up · 36 | 3

Core · 8 | 64

Challenge · 72 | 72 | 36

NAME __________
DATE __________
SCORE ____ / 11

Plotting Points

Across the hall, then up the stairs. Type coordinates like 3,5.

Warm-Up

6 ITEMS

In (3, 5), the number across	In (3, 5), the number up	The origin — type as a,b
4 across, 2 up — type as a,b	0 across, 6 up — type as a,b	In (7, 1), the first number

Core

GRADED · SHOW YOUR WORKING

1. 3 right of (2,5) — type as a,b
2. 2 up from (4,1) — type as a,b
3. Distance from (2,3) to (7,3)
4. Distance from (4,1) to (4,8)
5. Halfway between (0,0) and (8,0) — type as a,b

Challenge

NOT GRADED

C1. Halfway between (2,2) and (8,6) — type as a,b

C2. From (0,0) to (6,8) going only across and up — total steps

Shapes on the Grid

Plot the corners, then read off what you built.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up

6 ITEMS

Corners on a rectangle	Corners on a triangle	(1,1) to (5,1) — the length
(1,1) to (1,4) — the length	Sides on a square	(0,0) to (3,0) — the length

Core

GRADED · SHOW YOUR WORKING

1. (1,1), (1,5), (6,5), (6,1) — the area
2. That shape's perimeter
3. (2,2), (2,7), (5,7), (5,2) — the area
4. (0,0), (0,4), (4,4), (4,0) — the shape's name
5. The missing corner of a rectangle at (1,1),(1,4),(6,4) — type as a,b

Challenge

NOT GRADED

C1. (0,0), (8,0), (8,5), (0,5) — the area

C2. A square with corners (2,2) and (7,7) opposite — its area

Reading a Map

Coordinates are directions somebody else has to follow.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

From (0,0), move 3 across — type as a,b	From (3,0), move 2 up — type as a,b	From (5,5), move 1 across — type as a,b
From (2,2), move 2 up — type as a,b	Blocks from (0,0) to (4,0)	Blocks from (0,0) to (0,7)

Core GRADED · SHOW YOUR WORKING

1. From (1,2) to (6,2) then up 3 — type the end as a,b
2. Total blocks travelled on that route
3. A park at (3,4), a school at (9,4) — blocks apart
4. From (2,1) to (2,9) — blocks
5. From (0,0) to (5,5) across-and-up — blocks

Challenge NOT GRADED

C1. A route (1,1) to (7,1) to (7,6) — total blocks

C2. The shortest across-and-up route from (2,3) to (9,8) — blocks

Order Matters

(3,5) and (5,3) are different places. Prove it every time.

NAME __________

DATE __________

SCORE ____ / 11

Warm-Up

6 ITEMS

In (3,5), the across value	In (5,3), the across value	Are (3,5) and (5,3) the same — yes or no
In (0,4), the across value	In (4,0), the up value	A point on the bottom line has which value 0 — type across or up

Core

GRADED · SHOW YOUR WORKING

1. (2,6) and (6,2) — how far apart across
2. Those two points — how far apart up
3. A point with the same across and up as (5,5) — type as a,b
4. Points where across equals up, from (0,0) to (5,5) — how many
5. Swap (7,2) — type the new point as a,b

Challenge

NOT GRADED

C1. Points on the line where across equals up, from 0 to 10 — how many

C2. (1,4), (4,1), (1,1) — the area of that triangle

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

NAME __________
DATE __________
SCORE ____ / 7

Warm-Up

2 ITEMS

Perimeter of a 4 by 3 rectangle

Area of a 4 by 3 rectangle

Core

GRADED · SHOW YOUR WORKING

1. Area 48, one side 6 — the other
2. With 20 m of fence, the biggest area
3. An 8 by 6 with a 2 by 3 removed — area
4. Distance from (2,3) to (7,3)
5. (1,1), (1,5), (6,5), (6,1) — the area

Challenge

NOT GRADED

C1. A rectangle of perimeter 30 with whole sides — how many different ones

Week 3 Answers

Mission 06 · all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

3.1 · Plotting Points

OUT OF 11

Warm-Up · 3 | 5 | 0,0 | 4,2 | 0,6 | 7

Core · 5,5 | 4,3 | 5 | 7 | 4,0

Challenge · 5,4 | 14

3.2 · Shapes on the Grid

OUT OF 11

Warm-Up · 4 | 3 | 4 | 3 | 4 | 3

Core · 20 | 18 | 15 | square | 6,1

Challenge · 40 | 25

3.3 · Reading a Map

OUT OF 11

Warm-Up · 3,0 | 3,2 | 6,5 | 2,4 | 4 | 7

Core · 6,5 | 8 | 6 | 8 | 10

Challenge · 11 | 12

3.4 · Order Matters

OUT OF 11

Warm-Up · 3 | 5 | no | 0 | 0 | up

Core · 4 | 4 | 5,5 | 6 | 2,7

Challenge · 11 | 4.5

Fri · Mid-Unit Quiz

OUT OF 7

Warm-Up · 14 | 12

Core · 8 | 25 | 42 | 5 | 20

Challenge · 7

Lines of Symmetry

Fold it. If both halves land exactly, that fold is a line of symmetry.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up
6 ITEMS

Lines of symmetry in a square	In a rectangle that is not a square	In a circle — type the word
In an equilateral triangle	In the letter A	In the letter H

Core
GRADED · SHOW YOUR WORKING

1. In a regular pentagon ....
2. In a regular hexagon ....
3. In an isosceles triangle ....
4. In a scalene triangle ....
5. In a rhombus that is not a square ....

Challenge
NOT GRADED

C1. In a regular octagon

C2. In a regular polygon with 12 sides

Flips and Turns

Slide, flip, turn. The shape keeps its size and its angles.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

Degrees in a quarter turn	Degrees in a half turn	Degrees in a full turn
Quarter turns in a full turn	Degrees in three quarter turns	Half turns in a full turn

Core GRADED · SHOW YOUR WORKING

1. A square turned 90° looks the same — yes or no
2. Turns of 90° before a square returns to start
3. A rectangle looks the same after how many degrees
4. (2,3) flipped over the vertical line through 0 — the across value becomes
5. Slide (2,3) four right — type as a,b

Challenge NOT GRADED

C1. An equilateral triangle looks the same after how many degrees

C2. A regular hexagon — the smallest turn that leaves it unchanged, in degrees

Angles

Right, acute, obtuse — and what they add to.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up

6 ITEMS

Degrees in a right angle	Degrees on a straight line	Degrees round a point
An angle of 45° — type acute or obtuse	An angle of 120° — acute or obtuse	Angles in a triangle add to

Core

GRADED · SHOW YOUR WORKING

1. Two angles on a line, one is 60° — the other
2. A triangle with 90° and 30° — the third angle
3. Angles in a quadrilateral add to
4. A quadrilateral with three 90° angles — the fourth
5. Three angles round a point, two are 100° — the third

Challenge

NOT GRADED

C1. Each angle of an equilateral triangle

C2. Each angle of a regular hexagon

Classifying Shapes

Sides, angles, parallels. Name it from its properties.

NAME __________
DATE __________
SCORE ____ / 11

✦ Warm-Up 6 ITEMS

Sides on a pentagon	Sides on a hexagon	Sides on an octagon
Sides on a quadrilateral	Equal sides on an equilateral triangle	Sides on a triangle

◆ Core GRADED · SHOW YOUR WORKING

1. Is every square a rectangle — yes or no ....
2. Is every rectangle a square ....
3. Pairs of parallel sides in a parallelogram ....
4. Pairs of parallel sides in a trapezoid ....
5. A triangle with one 90° angle — type its name ....

★ Challenge NOT GRADED

C1. A rhombus with right angles is also called a

....

C2. A triangle with all sides different — type its name

....

NAME __________
DATE __________
SCORE ____ / 4

Tessellation

Which shapes tile a floor with no gaps? Test three and explain.

Warm-Up

2 ITEMS

Do squares tile with no gaps — yes or no

Do circles tile with no gaps

Core

GRADED · SHOW YOUR WORKING

1. Do equilateral triangles tile — yes or no
2. Do regular hexagons tile

Challenge

NOT GRADED

C1. Do regular pentagons tile — yes or no

C2. Angles must meet at a point adding to

C3. Hexagons at a point: 360 ÷ 120 — how many meet

TEACHER PAGE · ANSWER KEY

Week 4 Answers

Mission 06 · all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

4.1 · Lines of Symmetry

OUT OF 11

Warm-Up · 4 | 2 | infinite | 3 | 1 | 2

Core · 5 | 6 | 1 | 0 | 2

Challenge · 8 | 12

4.2 · Flips and Turns

OUT OF 11

Warm-Up · 90 | 180 | 360 | 4 | 270 | 2

Core · yes | 4 | 180 | -2 | 6,3

Challenge · 120 | 60

4.3 · Angles

OUT OF 11

Warm-Up · 90 | 180 | 360 | acute | obtuse | 180

Core · 120 | 60 | 360 | 90 | 160

Challenge · 60 | 120

4.4 · Classifying Shapes

OUT OF 11

Warm-Up · 5 | 6 | 8 | 4 | 3 | 3

Core · yes | no | 2 | 1 | right

Challenge · square | scalene

Fri · Tessellation

OUT OF 4

Warm-Up · yes | no

Core · yes | yes

Challenge · no | 360 | 3

Map a Planet

An invented world on the coordinate plane, with real distances.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

A harbour at (2,3) — the across value	A peak at (7,7) — the up value	Distance from (2,3) to (6,3)
Distance from (5,1) to (5,6)	The origin — type as a,b	3 across, 8 up — type as a,b

Core GRADED · SHOW YOUR WORKING

1. Sailing (1,1) to (9,1) then to (9,5) — total distance
2. A square island (2,2) to (8,8) — its area
3. That island's perimeter
4. Halfway between (0,0) and (10,4) — type as a,b
5. A route (0,0) to (4,0) to (4,7) — distance

Challenge NOT GRADED

C1. A rectangular sea from (1,1) to (11,6) — its area

C2. Ten landmarks each 3 units apart in a line — total length

Sailing Directions

Somebody else has to follow them. Ambiguity is the enemy.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

From (0,0) move 5 across — type as a,b	Then 3 up — type as a,b	Total distance travelled
From (4,4) move 2 up — type as a,b	From (4,4) move 2 across — type as a,b	Blocks from (0,0) to (3,4) across-and-up

Core GRADED · SHOW YOUR WORKING

1. (1,1) to (1,7) to (5,7) — total distance
2. (2,2) to (8,2) to (8,9) — total distance
3. A round trip (0,0) to (6,0) and back — distance
4. (3,3) to (3,10) — distance
5. A square patrol of side 5 — total distance

Challenge NOT GRADED

C1. A patrol (1,1),(7,1),(7,5),(1,5) and back to start — total distance

C2. The area enclosed by that patrol

Mission Review

Everything from five weeks, mixed together.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

Perimeter of a 5 by 5 square	Area of a 5 by 5 square	Degrees in a right angle
Lines of symmetry in a square	In (3,5), the up value	Angles in a triangle add to

Core GRADED · SHOW YOUR WORKING

1. Area 96 with one side 12 — the other
2. A triangle with 90° and 30° — the third angle
3. With 24 m of fence, the biggest area
4. (2,2), (2,7), (5,7), (5,2) — the area
5. Lines of symmetry in a regular hexagon

Challenge NOT GRADED

C1. A 10 by 10 with a 4 by 4 removed — area

C2. Each angle of a regular hexagon

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up

6 ITEMS

Perimeter of a 6 by 6 square	Area of a 8 by 3 rectangle	Degrees round a point
Sides on a hexagon	Distance from (2,3) to (7,3)	Lines of symmetry in a rectangle

Core

GRADED · SHOW YOUR WORKING

1. An 8 by 6 with a 2 by 3 removed — area
2. A square of perimeter 36 — one side
3. Two angles on a line, one is 60° — the other
4. (1,1), (1,5), (6,5), (6,1) — the perimeter
5. Is every square a rectangle — yes or no

Challenge

NOT GRADED

C1. Among rectangles with perimeter 40, the biggest area

C2. A regular hexagon — smallest turn leaving it unchanged, in degrees

Mission 06 Test

Twelve items plus the Big Question, answered out loud.

NAME __________
DATE __________
SCORE ____ / 10

Warm-Up 2 ITEMS

Perimeter of a 10 by 2 rectangle

Area of a 10 by 2 rectangle

Core GRADED · SHOW YOUR WORKING

1. Area of a 12 by 8 rectangle
2. A rectangle of perimeter 20, one side 6 — the other
3. With 20 m of fence, the biggest area
4. A 12 by 5 with a 3 by 5 removed — area
5. Distance from (4,1) to (4,8)
6. (1,1), (1,5), (6,5), (6,1) — the area
7. Lines of symmetry in a regular pentagon
8. A triangle with 90° and 30° — the third angle

Challenge NOT GRADED

C1. A 20 by 15 room with a 4 by 5 alcove added — total area

C2. A patrol (1,1),(7,1),(7,5),(1,5) — the area enclosed

TEACHER PAGE · ANSWER KEY

Week 5 Answers

Mission 06 · all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

5.1 · Map a Planet

OUT OF 11

Warm-Up · 2 | 7 | 4 | 5 | 0,0 | 3,8

Core · 12 | 36 | 24 | 5,2 | 11

Challenge · 50 | 27

5.2 · Sailing Directions

OUT OF 11

Warm-Up · 5,0 | 5,3 | 8 | 4,6 | 6,4 | 7

Core · 10 | 13 | 12 | 7 | 20

Challenge · 20 | 24

5.3 · Mission Review

OUT OF 11

Warm-Up · 20 | 25 | 90 | 4 | 5 | 180

Core · 8 | 60 | 36 | 15 | 6

Challenge · 84 | 120

Thu · Error Journal Sweep

OUT OF 11

Warm-Up · 24 | 24 | 360 | 6 | 5 | 2

Core · 42 | 9 | 120 | 18 | yes

Challenge · 100 | 60

Fri · Mission 06 Test

OUT OF 10

Warm-Up · 24 | 20

Core · 96 | 4 | 25 | 45 | 7 | 20 | 5 | 60

Challenge · 320 | 24